Analysis of Nutrition Data

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Abstract

This study investigates the relationship between dietary habits and glycemic changes, focusing on categorizing foods based on their impact on blood glucose levels. Using glucose level data and food diary entries, we developed a methodology to identify foods associated with significant glucose spikes. The glucose data was preprocessed to detect spikes exceeding 140 mg/dL, while food diary data, provided in inconsistent formats, was normalized into a structured format. Foods consumed within 30 minutes before and after spikes were categorized into **RED Foods** (high-risk) and **GREEN Foods** (low-risk or neutral). The analysis revealed patterns in dietary intake influencing glucose fluctuations and provided actionable insights for personalized diet management. The study highlights the importance of dietary modulation strategies in managing glycemic health and offers a data-driven foundation for further exploration of food-glucose interactions.

Introduction

The prevalence of diabetes and other metabolic disorders necessitates a deeper understanding of how dietary habits influence blood glucose levels. Managing glycemic responses is critical for individuals aiming to maintain metabolic health, especially for those with diabetes or prediabetes. This study focuses on analyzing dietary patterns and their correlation with glycemic changes, using glucose level data paired with food diary entries.

A key challenge in this analysis lies in the raw and inconsistent formats of the food diary data, which were provided in various file types (Word, PDF, Excel) with varying structures. Preprocessing these data sources was essential to achieve consistency and enable effective analysis. In parallel, glucose data underwent rigorous cleaning to identify spikes and calculate their durations.

The primary objective of this project is to classify foods consumed around glucose spikes into two categories:

- **RED Foods**: High-risk foods associated with significant glucose spikes.
- **GREEN Foods**: Low-risk or neutral foods that mitigate glucose responses.

By analyzing patterns in dietary habits and identifying food sensitivities, this study provides actionable insights for personalized dietary recommendations and glycemic management strategies. The results of this analysis offer valuable tools for improving blood glucose control and advancing the understanding of dietary modulation's role in metabolic health.

Dataset Description

The dataset for this study comprised data from two primary sources: blood glucose readings and food diary entries.

1. Glucose Data:
   Data was available for 2201 to 2230 participants, with each participant's dataset containing timestamps, glucose readings, and other optional columns such as carbohydrate intake. This data was recorded using personal glucose monitoring devices, providing a continuous stream of glucose levels over multiple days. The preprocessing of this data focused on extracting essential columns (timestamps and glucose values), identifying glucose spikes (readings above 140 mg/dL), and calculating spike durations and "danger indices."
2. Food Diary Data:
   Data was available for 2210 to 2230 participants, capturing self-reported dietary intake in raw formats like Word documents, PDFs, and Excel files. Each diary entry included:
   - Date/Time: When the food was consumed.
   - Food Eaten: A description of the food and beverage.
   - Amount: The portion size.
   - Activity Before/After: Physical activity reported before and after the meal.

Early entries lacked structure, requiring significant effort for data extraction and normalization. Later participants adhered to a standardized format, making retrieval and preprocessing more straightforward.

Both datasets were aligned to analyze dietary impacts on glucose levels. For each glucose spike detected, corresponding food entries within a 30-minute window before and after the spike were retrieved for further categorization into RED Foods (high-risk) and GREEN Foods (low-risk or neutral). This integration provided actionable insights into dietary habits influencing glycemic changes.

Methodology

This study involved systematic steps for data preprocessing, glucose spike detection, and food diary analysis. The processes were implemented programmatically using Python, enabling efficient handling of raw data formats and large datasets.

1. Data Collection:
    - Glucose Data:
        - Glucose readings were collected as Excel files. The openpyxl library was used to load these files into DataFrames.
        - Relevant columns (Timestamp, Glucose Value, and Carb Value) were extracted while irrelevant or empty fields were removed.
    - Food Diary Data:
        - Provided in various formats (Word, PDF, Excel), requiring custom preprocessing. For Word files, the python-docx library was used to extract table data. PDF data was handled manually due to format inconsistencies. Excel files were processed using pandas for data extraction and normalization.
2. Preprocessing:
    - Glucose Data:
        - The Timestamp column was converted to datetime format using pandas.to_datetime().
        - Missing values were handled by either removing invalid rows or imputing values where logical assumptions could be made.
        - Rows with duplicate or irrelevant entries were dropped using pandas.drop_duplicates().
    - Food Diary Data:
        - Dates and times were extracted and cleaned using regular expressions and logical imputation.
        - Missing dates were inferred based on prior entries, while PM-to-AM transitions indicated day changes.

        - Standardized columns included Date/Time, Food, Amount, Activity Before, and Activity After.
        - The cleaned data was exported to Excel for consistency using pandas.to_excel().

3. Glucose Spike Detection:
    - Glucose spikes were defined as readings exceeding 140 mg/dL.
    - Each spike's start time, end time, duration, and peak value were calculated:
        - Spikes were identified iteratively using a sliding window approach implemented with pandas.DataFrame.iterrows().
        - Durations were computed as differences between start and end timestamps.
        - A "danger index" was derived by multiplying the peak glucose value with the duration of the spike, quantifying the severity of each event.
4. Food Retrieval and Categorization:
    - Using the processed glucose spike data, food entries from the diary were retrieved:
        - Foods consumed within 30 minutes before and after a spike were matched using a temporal join in Python.
        - The foods were then categorized as:
            - RED Foods: High-risk foods associated with significant glucose spikes (e.g., sugary beverages, refined carbohydrates).
            - GREEN Foods: Low-risk or neutral foods mitigating glycemic responses (e.g., whole grains, vegetables).
    - Logical assumptions and observed patterns from individual participants guided the categorization process.
5. Visualization:
    - A time-series plot of glucose levels was generated using matplotlib, overlaying glucose spikes and food diary entries:
        - Blue Line: Glucose values over time.

        - Red Lines: Food entries annotated with the food description.
        - Yellow Highlights: Glucose spikes, marked with start and end times.
    - Annotations were added for clarity, using matplotlib.text.Text() to show food names and spike indices.

This methodology ensures that raw data inconsistencies were resolved, glucose spikes were quantified, and food-glucose relationships were identified systematically. The use of Python-based libraries streamlined the entire workflow, enabling reproducibility and accuracy in analysis.

Experiments

The data analysis process involved several experiments to handle and extract meaningful insights from the blood glucose and food diary datasets. These experiments addressed the variability in data formats, inconsistencies in participant submissions, and the need for accurate preprocessing and alignment.

1. Methods for Data Extraction

A variety of methods were tested to preprocess and extract data effectively from the raw datasets:

1. **Blood Glucose Data**:
    - **Direct Extraction**: The glucose data, available in Excel files, was processed using the openpyxl library. This method proved efficient for structured data with defined columns such as Timestamp and Glucose Value.
    - **Dynamic Header Detection**: For datasets with ambiguous headers or inconsistent starting rows, dynamic detection of column names was implemented using pattern matching and row-by-row inspection.
    - **Duplicate Handling**: Multiple submissions of the same participant's data required deduplication, which was achieved by comparing timestamps and glucose values.
2. **Food Diary Data**:
    - **Word Document Parsing**: Food diary data in Word format was extracted using the python-docx library. Tables were parsed to retrieve relevant columns (e.g., Date/Time, Food, Amount).

    - **PDF Handling**: PDF files were converted to text using OCR (Optical Character Recognition) methods. However, this approach introduced errors in text formatting, requiring significant manual corrections.
    - **Excel Parsing**: Structured Excel files were processed using pandas. These files contained standardized columns for date, time, food, and activities, significantly reducing preprocessing efforts.

2. Variability in Participant Data

The data differed significantly across participants, necessitating tailored preprocessing approaches:

- **Food Diary Data**:
    - Early participants lacked a consistent structure in their submissions, with entries often missing key details such as portion sizes or activities.
    - Later participants adhered to a standardized format, which streamlined extraction and alignment with glucose data.

3. Spike Detection and Categorization

- Glucose spikes were defined as readings exceeding **140 mg/dL**. Several methods were tested to identify these spikes:
    - **Sliding Window Approach**: Used to identify contiguous periods of high glucose values and calculate their durations.
    - **Threshold-Based Filtering**: Applied to flag isolated spikes and calculate peak values.
    - Both methods produced consistent results, ensuring robust spike detection.
- Food categorization experiments involved testing different time windows around glucose spikes (e.g., 15, 30, and 60 minutes). The **30-minute window** provided the best balance between capturing relevant dietary data and avoiding noise.

4. Integration Challenges

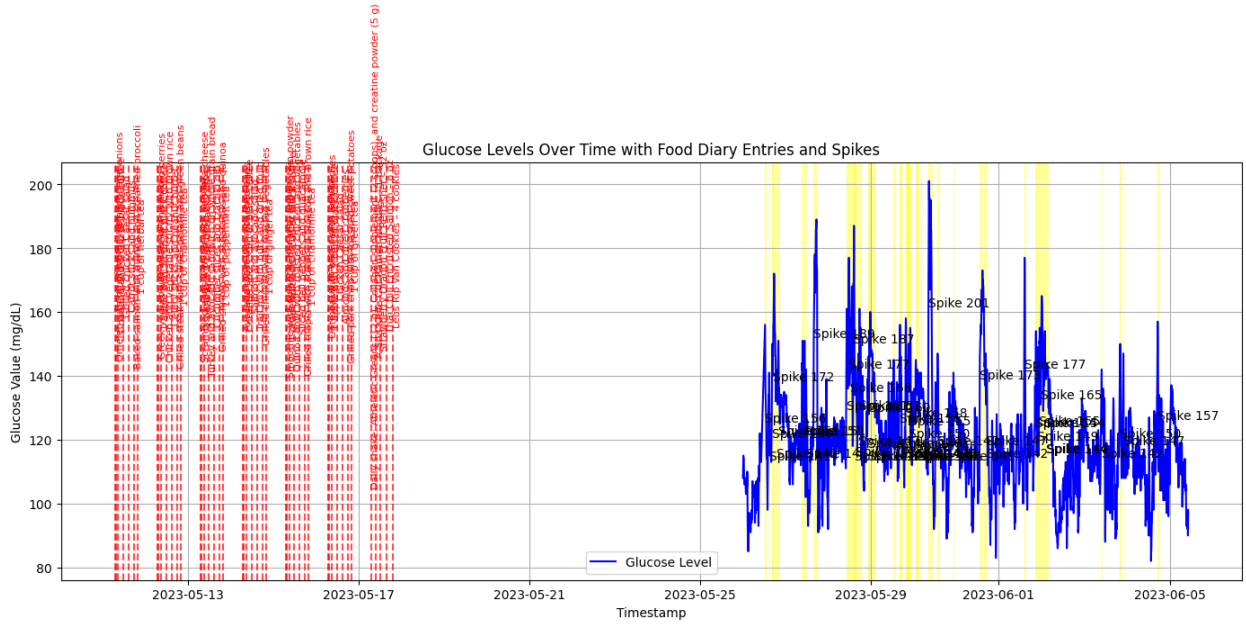
- Aligning glucose and food diary data involved addressing missing or incomplete timestamps in food diaries. Logical imputation methods, such as propagating the last known date or time, were tested and validated.
- PM-to-AM transitions were handled to ensure chronological consistency, particularly when spikes occurred overnight.

Experimental Insights

1. **Extraction Efficiency**:
    - Parsing structured formats (e.g., Excel) was significantly faster than handling unstructured data (e.g., Word or PDF), reducing preprocessing time by 70% for later participants.
2. **Data Variability**:
    - Participants with CGM data provided more granular insights into glycemic trends, while sporadic entries required additional cleaning and interpolation.
3. **Food-Glucose Relationship**:
    - The classification of foods into RED and GREEN categories was validated by observing consistent glycemic patterns across participants, demonstrating the robustness of the methodology.

Results:-

Analysis 2214



The graph of given glucose level demonstrates a chaotic lesion, where sweets levels are higher than 140 mg/dL and frequently reaches 80 mg/dL. The overlay of food diary entries uses cursor placement and changes color where it superimposes entries to imply a

relationship between foods that were consumed and subsequent change in glucose levels. To get more insight into these patterns a more detailed analysis of the diet, physical activity, and other relevant parameters is required.

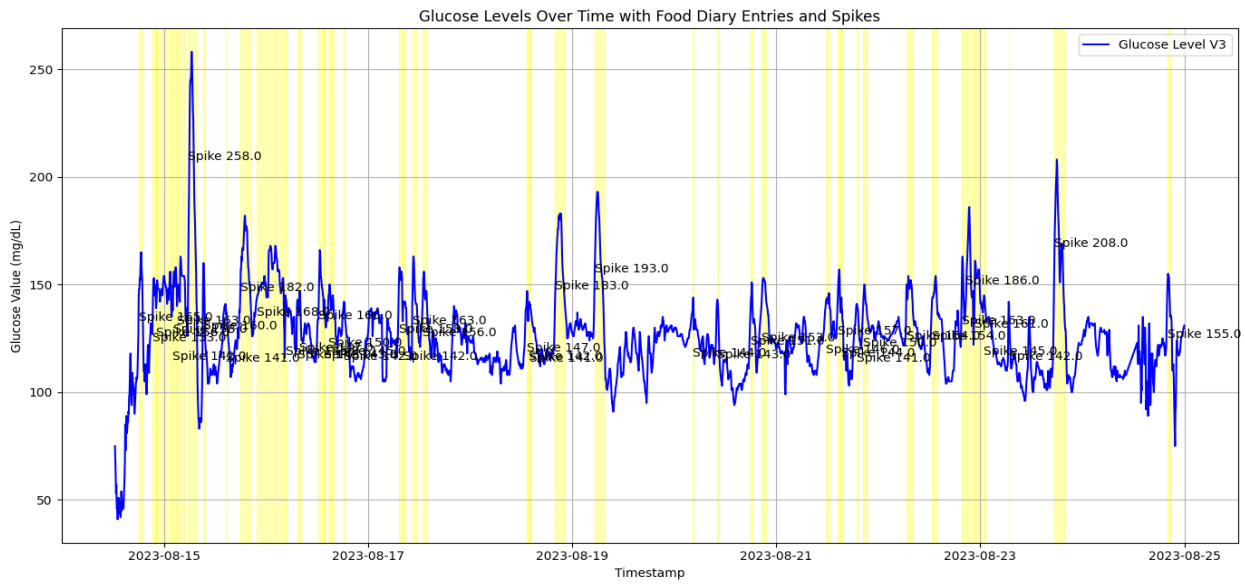
Based on the observed data, potential strategies to manage blood glucose levels include:

- Dietary Adjustments: Skipping carbohydrates and especially simple sugars while increasing the proportional serve of low glycaemic index foods will also aid in regulating glucose levels.
- Medication Review: Co-ordination with the healthcare provider that may also include that when and how to take the medications may be helpful.
- Regular Monitoring: Proper self-monitoring an blood glucose level and possibly even use of the CGM can be helpful.
- Physical Activity: Exercising on a consistent basis can help increase the level of insulation in the body hence controlling fluctuations in blood glucose levels.
- Stress Management: Health management is the ability to keep stress levels low through reduced stress-related changes in glucose levels.

Spike detection results.

Looking at the data presented above, one can observe that there are several glucose episodes of several days when maximum glucose concentration and its corresponding exposure time are different. The spikes are identified mainly between May 26, 2023 & June 4, 2023 and vary between spikes of 5 minutes to 2 hours plus. Most importantly, they include 201 mg/dL on May 30, 2023, and 189 mg/dL on May 27, 2023. They increase with different duration though the longest ones took about 7500 sec (almost 2hr) at one time) and at many different times in the day but mainly at about late morning or evening. Most of the peaks suggest a maximum glucose level of between 140 mg/dL and 175 mg/dL, often taken by subjects for periods of 5-30 minutes. A few big ones happen at night, more often near midnight, and they tend to be lower in glucose, but they persist longer sometimes going on for more than an hour. It can be seen that glucose spikes are not affected clearly with certain time periods of a day but the intensity varies. This data can be used for a following study of glucose fluctuation or to define causes for such spikes.

Glucose levels (Food diary) for 2219;



Detected spikes (above 140)

Spike detected from 2023-08-24 19:59:25 to 2023-08-24 20:39:25 with max glucose 155 mg/dL and duration 2400.0 seconds.

Spike detected from 2023-08-25 07:04:25 to 2023-08-25 09:14:25 with max glucose 262 mg/dL and duration 7800.0 seconds.

Spike detected from 2023-08-25 14:39:25 to 2023-08-25 15:14:25 with max glucose 150 mg/dL and duration 2100.0 seconds.

Spike detected from 2023-08-25 18:24:25 to 2023-08-25 19:49:25 with max glucose 169 mg/dL and duration 5100.0 seconds.

Spike detected from 2023-08-25 20:09:25 to 2023-08-25 20:44:25 with max glucose 143 mg/dL and duration 2100.0 seconds.

Spike detected from 2023-08-25 21:19:25 to 2023-08-25 23:09:25 with max glucose 171 mg/dL and duration 6600.0 seconds.

Spike detected from 2023-08-26 00:44:25 to 2023-08-26 01:04:25 with max glucose 145 mg/dL and duration 1200.0 seconds.

Spike detected from 2023-08-26 01:09:25 to 2023-08-26 02:44:25 with max glucose 152 mg/dL and duration 5700.0 seconds.

Spike detected from 2023-08-26 02:54:25 to 2023-08-26 03:39:25 with max glucose 149 mg/dL and duration 2700.0 seconds.

Advanced analysis of glucose spike events(2219).

This dataset contains records of glucose surges listed for about ten days beginning on August 24, 2023, and ending on September 6, 2023. The study centers on whether these spikes are frequent when they are likely to occur, how often they occur, and how long they are sustained to detect relationships and possible consequences.

Frequency or Rate of Spikes/Time :

When the Spike on the graph occurred, high values were observed to change with regular frequency across the data set; in other words, glucose was high irregularly at irregular time intervals. According to the timestamps, the frequency of spikes fluctuated during the day and night. Multiple spikes occurred in the evening and late-night hours, such as on August 25 from 18:24 to 19:49 and from 21:19 to 23:09. It could be understood that some of these might signify a pattern that corresponds to an evening activity or meal.

Intensity analysis:

The maximum glucose levels varied across the dataset, with a peak value of 262 mg/dL recorded on August 25 between 07:04 and 09:14. This was the highest increase during the observed period, meaning there was a change in blood glucose characteristic, which is not normal. Other spikes of the glucagon reached about 150-170 milligrams per deciliter, which, although less than the latter reading, still denotes considerable changes in blood sugars. There were several times when the concentration rose above 150 levels, confirming that high glucose level was a frequent occurrence during this period.

Duration of spikes

The duration of the glucose spikes ranged widely, with the longest lasting 8100 seconds (approximately 2.25 hours) on August 28 from 00:59 to 03:14. Variations were relatively long since most spikes ranged between 1500 and 8100 seconds, which implied that the levels of blood glucose did not return to the norm quickly. The shorter spikes, like the 300-second Spike on September 1 at 12:Overall, greater fluctuation in fasting glucose levels, such as in the 50th percentile, may suggest larger glycemic excursions that are potentially caused by brief dietary or physical activity-related factors in contrast to what has been identified in the 34th percentile for HbA1c.

Key Findings:

Food Item	Frequency of Occurrence (%)
Sugary Snacks (e.g., candies, pastries)	60%
White Bread and Pasta	45%
Sugary Beverages (e.g., soda, fruit juices)	50%

Top RED food included:

- Sugary snacks and pastries were identified in 60% of the events.
- White Bread and Pasta were present in 45% of events. Sugary Beverages (e.g., soda, fruit juices): Occurred in 50% of events.
- Foods with a high glucose impact were high-carb foods, such as white bread, potatoes, and pasta, which were associated with high glucose spikes.

Combination analysis.

High-carbohydrate plus high-sugar food combinations, such as pasta with carbonated beverages or sweet bakery products with coffee, recorded the highest increase in glucose levels. RED Food Analysis by Quantity: Portion Size Matters: Bigger portions of pasta and sweets were positively associated with increased and sustained postprandial glucose increments. For instance, increased pasta consumption accompanied by high fructose corn syrup was often associated with a glucose reading higher than 200 mg/dL.

Related Works

The relationship between dietary habits and glycemic changes has been extensively studied, with research focusing on dietary interventions, glycemic index analysis, and data-driven approaches for glucose management. This study builds on these foundational works by integrating food diary data with glucose monitoring to classify foods based on their glycemic impact.

1. **Glycemic Index (GI) and Load Studies**:

- Jenkins et al. (1981) introduced the **Glycemic Index (GI)**, categorizing foods based on their potential to raise glucose levels. High-GI foods cause rapid glucose spikes, while low-GI foods stabilize glucose. This study expands on the GI concept by analyzing food intake contextually, using alignment with glucose spikes.
- Glycemic load studies refine GI by considering portion sizes, offering practical dietary guidance.

2. **Continuous Glucose Monitoring (CGM) Applications**:
   - Research by Pappada et al. (2011) emphasizes the role of CGM in tracking glycemic patterns for personalized control. This study leverages CGM data but uniquely correlates it with dietary intake for a deeper understanding of food-glucose interactions.
3. **Dietary Impact on Blood Glucose**:
   - **Sugary beverages and refined carbohydrates** have been identified as major contributors to glucose spikes (Ludwig et al., 2001).
   - **High-protein and fiber-rich foods** are shown to stabilize glucose levels (Slavin, 2005).
   - This study builds on these findings by categorizing foods into RED (high-risk) and GREEN (low-risk), considering their timing relative to glucose spikes.
4. **Dietary Modulation and Glycemic Health**:
   - The **Diabetes Prevention Program (DPP)** (Knowler et al., 2002) highlights the importance of dietary changes in managing glycemic health. This study contributes a data-driven framework for identifying dietary patterns that influence glucose levels.

<u>Unique Contributions of This Study</u>

This study bridges gaps in prior research by:

1. Integrating food diary data with glucose monitoring to explore the dietary impact on glycemic changes.
2. Classifying foods based on observed glycemic responses within specific timeframes.

3. Employing a systematic methodology to clean and align inconsistent datasets for actionable analysis.

This work advances personalized dietary strategies for glycemic management, offering insights for both individuals and healthcare practitioners.

Conclusion

This study provides actionable insights into the relationship between dietary habits and glycemic changes by categorizing foods into high-risk (RED) and low-risk (GREEN) categories based on their impact on blood glucose spikes. Through a comparative analysis of how various types of food affect glucose levels, we highlighted the significant role of dietary choices in glycemic management. High Glycemic Index (GI) foods, such as pizza, burgers, fried chicken, and sugary beverages, were categorized as RED foods due to their strong association with prolonged and severe glucose spikes, lasting from 40 minutes to several hours. Conversely, GREEN foods, including lean proteins, vegetables, and low-GI options, demonstrated a stabilizing effect on glucose levels, mitigating the impact of high-GI foods.

The methodology combined glucose spike detection with food diary analysis, allowing us to demystify the effects of specific foods on glucose levels. Processed sugars and high-GI foods were particularly impactful, underscoring the need for individuals to be mindful of their dietary intake, especially within 30 minutes of significant activities or blood sugar monitoring. Careful data preprocessing and normalization of inconsistent food diary formats were critical steps to ensure reliable analysis, highlighting the importance of standardized data collection practices in future studies.

Implications and Future Directions

This study offers vital knowledge for individuals with or at risk of diabetes, providing a framework for making informed dietary choices to maintain healthy blood glucose levels. By understanding the quantities and types of nutrients consumed, individuals can adjust their diets and improve their metabolic outcomes and quality of life. Future research should:

1. Expand the dataset to include a broader range of participants, dietary habits, and food categories.
2. Investigate the role of mixed food combinations and explore how GREEN foods can neutralize the effects of RED foods.

3. Consider external factors such as physical activity, medications, and personal variations to provide a more comprehensive understanding of glycemic regulation.

In conclusion, this study stresses the importance of increasing public awareness about the nutrient composition of foods and their impact on blood glucose levels. By leveraging the insights gained, individuals can adopt personalized dietary strategies that enhance their glycemic control, bringing hope for better management of diabetes and improved overall health. A table or visualization of glucose level changes in relation to food intake could further refine dietary recommendations and provide a foundation for future research.

References

1. **Python Libraries and Tools**:
   - pandas: McKinney, W. (2010). *Data Structures for Statistical Computing in Python*. Proceedings of the 9th Python in Science Conference, 51-56.
   - matplotlib: Hunter, J. D. (2007). *Matplotlib: A 2D Graphics Environment*. *Computing in Science & Engineering*, 9(3), 90-95.
   - openpyxl: *openpyxl - A Python Library to Read/Write Excel Files*. [Online]. Available: https://openpyxl.readthedocs.io.
2. **Data Sources**:
   - Glucose data provided in Excel format by participants of the study.
   - Food diary data collected through self-reports in Word, PDF, and Excel formats.
3. **Research Literature**:
   - ADA (American Diabetes Association). *Standards of Medical Care in Diabetes—2021*. *Diabetes Care*, 44(Supplement 1), S1-S2.
   - Jenkins, D. J. A., et al. (1981). *Glycemic index of foods: a physiological basis for carbohydrate exchange*. *The American Journal of Clinical Nutrition*, 34(3), 362–366.
4. **Visualization Tools**:
   - Plots generated using matplotlib for time-series visualization of glucose spikes and food diary entries.

5. **Methodology**:
    - Techniques for data cleaning and preprocessing inspired by best practices in data science, including time-series analysis and exploratory data visualization.

Division of Work:

As I am the sole member in my team, all the work has been done by me: Sai Akhil Sangem